Tutorial 1 — Market Microstructure

LOB Features & Price Prediction

AI for Finance — M2 Paris-Saclay / CentraleSupélec

Instructions. Submit a single Python notebook (`.ipynb`) with your code, results, and written answers.

Timeline:
- **In class (1.5h):** Work on Parts 1–3. Leave the session with data loaded, features computed, and analytical baselines evaluated.
- **At home:** Complete Parts 4–5 (ML models and analysis).
- **Deadline: Monday 13 April, 23h59.**

Rules:
- **No data leakage.** Walk-forward (temporal) validation only. Future information leaking into training = strong penalty.
- **Show your work.** Explain your choices.
- **Interpret your results.** Raw numbers without commentary will not receive full marks.

Data

You will use limit order book (LOB) data. Several options:

(a) **FI-2010** — the standard academic benchmark for LOB prediction. 5 Finnish stocks, 10 trading days, 10 levels. Freely available:
`https://etsin.fairdata.fi/dataset/73eb48d7-4dbc-4a10-a52a-da745b47a649`
Reference: Ntakaris et al. (2018), *arXiv:1705.03233*.

(b) **LOBSTER** — reconstructed NASDAQ order books. Free sample data at `https://lobsterdata.com/info/DataSamples.php` (e.g. AAPL, AMZN, one day).

(c) Any other LOB dataset you have access to.

You need, at minimum: prices and volumes at K levels on both sides (bid and ask), for a sequence of LOB snapshots over time.

The raw LOB snapshot at depth K:

$$\mathbf{x}_t = \left(p_a^{(1)}, V_a^{(1)}, p_b^{(1)}, V_b^{(1)}, \; \dots \; , p_a^{(K)}, V_a^{(K)}, p_b^{(K)}, V_b^{(K)}\right) \in \mathbb{R}^{4K}$$

1 Data Exploration

Question 1.1

Compute and plot the **mid-price** $p_m(t) = \frac{1}{2}(p_a^{(1)} + p_b^{(1)})$ over time. Is it stationary?

Question 1.2

Compute and plot the **spread** $s(t) = p_a^{(1)} - p_b^{(1)}$. Report the mean, median, standard deviation, and the **exact timestamp** (or index) where the spread is largest. What is happening at that moment?

Question 1.3

Plot the LOB snapshot (standing volume per price level, bid and ask sides) for a few timestamps of your choice.

Question 1.4

Compute the mid-price returns $r_t = p_m(t) - p_m(t-1)$. Plot the histogram. Compute the kurtosis. Are returns Gaussian?

Report the **5 largest absolute returns** in your dataset (index, value, and what the LOB looked like at those moments). Are these genuine price moves or data artefacts?

2 Feature Engineering

Prediction target

Define the label:

$$y_t = \operatorname{sign}\big(p_m(t+\Delta t) - p_m(t)\big) \in \{-1, 0, +1\}$$

Use $\Delta t = 10$ (10 events ahead). You may also experiment with other horizons.

Question 2.0

Compute y_t. What is the class distribution? If one class dominates, explain how you handle this.

In Questions 2.1–2.5, add features ***one at a time****. For each: compute the feature, visualise its relationship with y_t, and evaluate how well it alone predicts y_t using a logistic regression on a temporal train/test split (e.g. first 80 % / last 20 %).*

Question 2.1

Order Imbalance (level 1): $\mathrm{OI}_t^{(1)} = \dfrac{V_b^{(1)} - V_a^{(1)}}{V_b^{(1)} + V_a^{(1)}}$

Report accuracy with this single feature. Is it better than random?

Question 2.2

Order Imbalance (levels 1–5): $\mathrm{OI}_t^{(k)} = \dfrac{V_b^{(k)} - V_a^{(k)}}{V_b^{(k)} + V_a^{(k)}}$ for $k = 1, \ldots, 5$.

Train a logistic regression using all 5 OI features. Do deeper levels add predictive power?

Question 2.3

Spread and volume-weighted mid-price deviation:

$$s_t = p_a^{(1)} - p_b^{(1)}, \qquad \delta_t = p_w(t) - p_m(t) \quad \text{where} \quad p_w = \frac{p_b^{(1)} V_a^{(1)} + p_a^{(1)} V_b^{(1)}}{V_b^{(1)} + V_a^{(1)}}$$

Add these to the OI features. Improvement?

Question 2.4

Lagged mid-price returns: $r_{t-1}, r_{t-2}, \ldots, r_{t-5}$.
Add them. Is there momentum or mean-reversion at the tick scale?

Question 2.5

Summary table. Present a table showing accuracy as features are added incrementally. Which features contribute the most? Which are redundant?

3 Analytical Baseline: No Machine Learning

Before using ML, build predictors based purely on microstructure theory.

Question 3.1

Parameter-free predictor. The volume-weighted mid-price p_w is a better estimate of the fair price than p_m. If $p_w > p_m$, predict up; otherwise, predict down:

$$\hat{y}_t = \operatorname{sign}(p_w(t) - p_m(t))$$

This requires **no training**. Evaluate its accuracy on the test set.

Question 3.2

Kyle-style linear model. Estimate by OLS on the training set:

$$\Delta p_m(t + \Delta t) = \alpha + \lambda \cdot \mathrm{OI}_t^{(1)} + \varepsilon_t$$

(a) What is the sign of $\hat{\lambda}$? Is this consistent with Kyle (1985)?
(b) Convert to directional predictions: $\hat{y}_t = \operatorname{sign}(\hat{\alpha} + \hat{\lambda} \cdot \mathrm{OI}_t^{(1)})$. Report test accuracy.
(c) Report R^2. Is it low or high? Interpret the results.

Question 3.3

Compare the two analytical models with the best logistic regression from Part 2. Are the analytical models competitive? Why or why not?

4 Machine Learning Models

Use your full feature set from Part 2. All models must be evaluated on the **same temporal test set**.

Question 4.1

Logistic Regression. Train on the training set, evaluate on test. Report accuracy, confusion matrix, and per-class precision/recall. This is your ML baseline.

Question 4.2

Gradient Boosting (XGBoost, LightGBM, or scikit-learn). Tune at least 2 hyperparameters (e.g. number of trees, max depth, learning rate). Report accuracy and feature importances.

Question 4.3

Multi-Layer Perceptron (MLP). Train a feedforward neural network with at least 2 hidden layers. Experiment with architecture (layer sizes, dropout, activation). Report accuracy.

Question 4.4

Recurrent model (LSTM or GRU). Instead of a single snapshot, feed **sequences** of T consecutive snapshots (e.g. $T = 50$) into an LSTM or GRU.

(a) Describe your architecture (layers, hidden size, sequence length T).
(b) Report accuracy. Does temporal information help compared to the MLP?

Question 4.5 (Bonus)

Walk-forward validation. Pick your best model. Implement walk-forward validation: split data into sequential blocks, train on all previous blocks, test on the next. Report accuracy per block and average. Is performance stable over time?

5 Analysis & Interpretation

Question 5.1

Final comparison. Create a summary table:

Model	Type	Test Accuracy
$\mathrm{sign}(p_w - p_m)$	analytical (no parameters)	?
OLS: $\Delta p = \alpha + \lambda \cdot \mathrm{OI}$	analytical (2 parameters)	?
Logistic Regression	ML (linear)	?
Gradient Boosting	ML (tree-based)	?
MLP	ML (neural network)	?
LSTM / GRU	ML (recurrent)	?

Which model wins? Is the gap between analytical and ML models large or small?

Question 5.2

Answer the following:

(a) Which single feature is the most predictive? Why does this make economic sense?

(b) Does the analytical model perform surprisingly well compared to ML? What does this tell you about the nature of the signal in LOB data?

(c) What are the main limitations of predicting mid-price direction at the tick level? (Consider: transaction costs, spread, class imbalance, non-stationarity.)

Question 5.3

Critical discussion. If you had more time and data, what would you try next? Discuss at least two ideas among: CNN on raw LOB snapshots (DeepLOB), transformers, different prediction horizons, multi-stock models, trade-level features, impact of non-stationarity.